

Hyunju Kim

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INTERESTS

My research investigates how *embodied representations*, including *telepresence robots* and *XR-based avatars*, enable and constrain collaboration across distance, space, and media. I focus on **proxy-mediated collaboration**, where embodied systems act as proxies for remote users in physical, virtual, and hybrid environments.

While prior XR research has demonstrated rich forms of interaction in virtual spaces, these insights often fail to translate into real-world collaborative settings. To address this gap, my work examines how spatial communication, coordination, and collaborative roles are shaped when interaction is mediated through embodied systems. Through system development and empirical studies, my research aims to inform the design of future collaborative technologies that meaningfully connect remote users with real-world environments.

RESEARCH EXPERIENCE

Cornell University

Aug 2022 – Present

– Robotic Telepresence for Remote Collaboration:

- Designing a VR-mediated telepresence navigation system that explores legibility of navigational intent and agency during side-by-side remote co-navigation with a mobile robot. (Ongoing project; [Demo video](#))
- Reframing Wizard-of-Oz as a method for designing and evaluating proxy-mediated interaction. (Under review)
- Designed and evaluated robotic and avatar-based telepresence systems to examine how embodiment and task structure shape collaboration roles, interaction strategies, and expectations. (Under external review for *CSCW 2026*; [Demo video](#))
- Designed and evaluated *VRoxy*, a VR-driven robotic proxy system for wide-area remote collaboration from small offices. (*UIST 2023*; [Demo video](#))
- Contributed to data analysis for *ReMotion*, an automatic robotic proxy that supports remote collaboration in open-space design activities. (*CHI 2023*; [Demo video](#))

– XR System for Reflective Collaboration:

- Developed an XR platform that transforms 2D rehearsal videos into interactive 3D avatars for embodied annotation, pose-based search, and choreographic revision. (*CHI 2026*; [Demo video](#))
- Awarded the [2024 PiTech Fellowship](#) in collaboration with the Martha Graham Dance Company for exploring 3D choreography archiving.

– Spatial Communication in Asymmetric XR Environments:

- Contributed to user study design and analysis for [MRTransformer](#) and [AvatarPilot](#), exploring non-verbal behavior transformation and motion interpolation for mixed-reality avatars.

AI & Robot Institution, KIST

May 2020 – Feb 2022

- Developed and evaluated a multi-sensory feedback system to support collaborative manipulation tasks in VR, enabling users to maintain dual awareness of shared actions. (*IJHCI*, 2023)
- Proposed a human-centered scheduling model for multi-agent environments, advancing dynamic task allocation in human–robot interaction. (*Applied Sciences*, 2022)

KAIST

Mar 2017 – Feb 2019

- Developed *RealityBrush*, an AR authoring system that captures and utilizes kinetic properties of everyday objects by combining an AR headset with a custom sensing module. ([Multimedia Tools Appl.](#); [Demo video](#))
- Contributed to study quantifying gender representation bias in commercial films through visual analysis.
- Evaluated and compared input devices to optimize performance in gaming contexts.

PROJECTS

Art+Tech Exhibit (A.D. White House)

Dec 2024

- Developed an **XR installation** where audience wearing VR headset could explore 3D archives of Martha Graham dancers.
- Presented at the Art+Tech Exhibition ([Cornell Chronicle](#)).

Gender Representation Analysis in Korean Films (Bechdel Day 2021)

Jul–Sep 2021

- Analyzed gender depiction across 2020 commercial and independent films by quantifying visual gender bias in film scenes using frame-by-frame analysis ([The Korea Times](#)).

Quiet Standing: Creative Code Exhibition (ArtCenter Nabi)

Sep–Dec 2019

- Developed an art project visualizing human existence through postural sway by measuring subtle body movement during quiet standing.
- Exhibited publicly at [ArtCenter Nabi](#), Seoul.

WORK EXPERIENCE

Research Intern, Korea Institute of Science & Technology (KIST)

May 2020 – Feb 2022

Developed VR collaboration and human–robot interaction systems under Dr. Jung-Min Park.

EDUCATION

2022 – Present Ph.D. in Information Science, **Cornell University**, Ithaca, NY

Advisor: Prof. François Guimbretière

2017 – 2019 M.Eng. in Culture Technology, **KAIST**, Daejeon, South Korea

Advisor: Prof. Byungjoo Lee;

Thesis: “FDPipette: Tracking Local Force and Deformation on Surrounding Objects to Augment Kinetic Reality.” ([Paper](#), [Demo video](#))

2013 – 2017 B.Sc. in Electrical Engineering, **Ewha Womans University**, Seoul, South Korea

Advisor: Prof. Jeongtae Kim;

Project: “Indoor Parking Navigation System.”

PUBLICATIONS

Wang, Cheng Yao, Hyunju Kim, et al. (Oct. 2024). “MRTransformer: Transforming Avatar Non-verbal Behavior for Remote MR Collaboration in Incongruent Spaces”. In: *Proceedings of the IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct)*, pp. 545–548. URL: <https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=10765238>.

- Wang, Cheng Yao, Eyal Ofek, et al. (Oct. 2024). “AvatarPilot: Decoupling One-to-One Motions from their Semantics with Weighted Interpolations”. In: *Proceedings of the IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct)*, pp. 588–591. URL: <https://www.microsoft.com/en-us/research/publication/avatarpilot-decoupling-one-to-one-motions-from-their-semantics-with-weighted-interpolations/>.
- Kim, Hyunju and Jung-Min Park (2023). “Providing Dual Awareness using Multimodal Cues for Collaborative Manipulation in Virtual Environments”. In: *International Journal of Human-Computer Interaction* 39.15, pp. 1–15. URL: <https://doi.org/10.1080/10447318.2023.2198965>.
- Kim, Hyunju, Yan Tao, et al. (Apr. 2023). “Comparing the Impact of Professional and Automatic Closed Captions on Video-Watching Experience”. In: *Extended Abstracts of the ACM Conference on Human Factors in Computing Systems (CHI)*, pp. 1–6. URL: <https://dl.acm.org/doi/full/10.1145/3544549.3585634>.
- Sakashita, Mose, Hyunju Kim, et al. (Oct. 2023). “VRoxy: Wide-Area Collaboration From an Office Using a VR-Driven Robotic Proxy”. In: *Proceedings of the ACM Symposium on User Interface Software and Technology (UIST)*, pp. 1–13. URL: <https://dl.acm.org/doi/10.1145/3586183.3606743>.
- Sakashita, Mose, Ruidong Zhang, et al. (Apr. 2023). “ReMotion: Supporting Remote Collaboration in Open Space with Automatic Robotic Embodiment”. In: *Proceedings of the ACM Conference on Human Factors in Computing Systems (CHI)*, pp. 1–14. URL: <https://dl.acm.org/doi/10.1145/3544548.3581527>.
- Jung, Yunseo et al. (2022). “Human-centered Dynamic Service Scheduling Approach in Multi-Agent Environments”. In: *Applied Sciences* 12.21, p. 10850. URL: <https://doi.org/10.3390/app122110850>.
- Kim, Hyunju, Sanghwa Hong, et al. (2021). “RealityBrush: An AR Authoring System that Captures and Utilizes Kinetic Properties of Everyday Objects”. In: *Multimedia Tools and Applications* 80, pp. 31135–31158. URL: <https://doi.org/10.1007/s11042-021-10769-4>.
- Kim, Hyunju et al. (2018). “A Study on the Effect of Inter-Key Spacing on Typing Performance”. In: *Proceedings of HCI Korea*, pp. 394–398. URL: <https://www.dbpia.co.kr/journal/articleDetail?nodeId=NODE07445025>.
- Park, Eunji et al. (Apr. 2018). “Button++: Designing Risk-aware Smart Buttons”. In: *Extended Abstracts of the ACM Conference on Human Factors in Computing Systems (CHI)*, pp. 1–6. URL: <https://dl.acm.org/doi/10.1145/3170427.3188621>.

AWARDS & ACHIEVEMENTS

PiTech Fellowship (\$14,000)	2024
KAIST Support Scholarship (\$32,000)	2017–2018
KAIST Graduate Student Fund (\$5,600)	2017–2018
SIGCHI Student Travel Grant (\$1,800)	2018
Excellence Award, Ewha Portfolio Competition	2017
First Prize, Continental Automotive Innovation Contest (\$1,700)	2016
First Prize, Ewha Capstone Design Competition (\$1,000)	2016
Dean’s List, Ewha Womans University	2014, 2015

TEACHING EXPERIENCE

- 2023 – Present **Cornell University**, Ithaca, NY
INFO 6520: HCI Design Studio;
INFO 5321: Rapid Prototyping and Physical Computing;
INFO 3450: Human–Computer Interaction Design;
INFO 2950: Introduction to Data Science
- 2017 – 2018 **KAIST**, Daejeon, South Korea
GCT 700: Research Methodology of Human Behavioral Science;
CTP 471: Social Networks Theory

SKILLS

- Programming C/C++, C#, Python, JavaScript, Arduino, Raspberry Pi, MATLAB, R, SPSS
Languages Korean (native), English (fluent)